

Notes: Greenwald & Guren (AER, 2025)

Do Credit Conditions Move House Prices?

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1 What this paper is doing

- **Question.** When credit conditions loosen, do we mainly observe:

1. a *tenure reallocation*: more owner-occupiers (higher HO), or
2. an *asset-price revaluation*: higher house prices relative to rents (higher p/q)?

- **Key mechanism.** There is a (local) **tenure supply curve** mapping the homeownership rate to the price–rent ratio:

$$\log(p/q) = S(\text{HO}), \quad S'(\text{HO}) \geq 0.$$

A credit easing shifts tenure *demand*. Whether equilibrium mainly moves “up” (prices) or “right” (homeownership) is governed by the slope $S'(\text{HO})$.

- **Empirical strategy.** Estimate the relative elasticities of $\log(p/q)$ and HO to plausibly exogenous credit-supply shocks.

- If $\partial \text{HO} / \partial Z$ is small but $\partial \log(p/q) / \partial Z$ is big, then $S'(\text{HO})$ must be large (steep tenure supply = segmentation).
- The paper reports the inverse ratio

$$\kappa \equiv \frac{\partial \text{HO} / \partial Z}{\partial \log(p/q) / \partial Z},$$

where small κ means high segmentation.

- **Structural model.** A dynamic housing model with borrowers (credit constrained), landlords (hold rental stock), and savers (marginal investors). A single parameter—dispersion in rental suitability across properties—controls tenure supply steepness.
- **Main result.** Empirically, credit easing raises p/q a lot and HO little. In the calibrated structural model, this implies that credit conditions can move house prices substantially even if homeownership barely changes.

2 All figures and tables (embedded) + interpretation

2.1 Figure 1: National evidence in the $(\text{HO}, p/q)$ plane

What we take from it.

- The 1998–2006 boom features a *large* rise in p/q and only a *modest* rise in HO.
- If credit mainly turned renters into owners, I would expect a stronger rightward move in HO.
- The observed pattern is more consistent with a steep tenure supply curve: demand pressure shows up in prices.

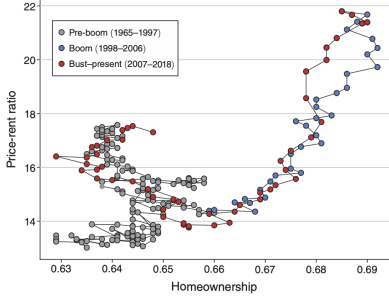


FIGURE 1. PRICE-RENT RATIO VERSUS HOMEOWNERSHIP RATE

Notes: The figure displays national data at the quarterly frequency. The price-rent ratio is obtained from the Bureau of Economic Analysis (BEA) and Financial Accounts of the United States, as the ratio of the value of residential housing owned by households (Federal Reserve Board of Governors 2021) to the value of owner-occupied housing services (US Bureau of Economic Analysis 2022). We use the method of Chow and Lin (1971) to convert owner-occupied housing services from annual to quarterly (see Supplemental Appendix B.5). The homeownership rate is obtained from the census (US Census Bureau 2022a).

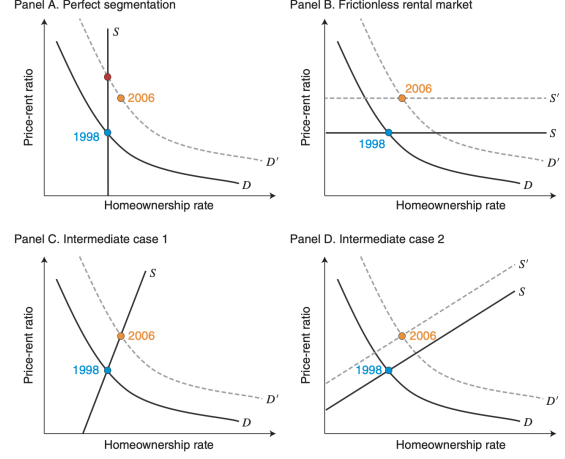


FIGURE 2. SUPPLY AND DEMAND INTUITION

2.2 Figure 2: Supply–demand intuition

Interpretation of the four panels.

- Panel A (perfect segmentation): supply is vertical, so credit-demand shifts move p/q but not HO.
- Panel B (no segmentation): supply is flat, so credit-demand shifts move HO but not p/q .
- Panels C–D: intermediate cases; both p/q and HO respond.

2.3 Figure 3: LS (CLL-based) credit shock — local projections

What Figure 3 shows.

- A credit easing increases p/q strongly and persistently.
- The HO response is small (often statistically indistinguishable from zero).
- The inverse ratio $\kappa_k = \beta_k^{\text{HO}} / \beta_k^{\log(p/q)}$ is near zero, implying a steep tenure supply curve.

2.4 Figure 4: DK (APL preemption) credit shock — event study

What Figure 4 shows.

- Post-2003, exposed areas experience a large rise in prices / p/q .
- Homeownership barely moves.
- Again: evidence for segmented tenure supply.

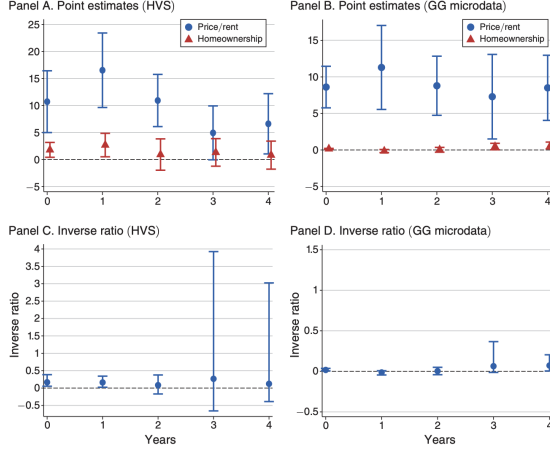


FIGURE 3. LS INSTRUMENT LP IMPULSE RESPONSES

Notes: The figure shows panel local projection estimates of the response of the indicated outcomes to the LS instrument $ShareNearCLL_{i,t-1} \times \%ChangeInCLL_t$ as estimated using equation (1). Control variables include $ShareNearCLL_{i,t-1}$ and its lag and lags of the instrument and outcome variables, and regressions are weighted by 2000 population. Panels A and B show the price/rent and homeownership rate for the HVS and GG homeownership rates, respectively, with standard errors clustered by CBSA. Panels C and D show the inverse ratio $\beta_{i,t}^{RR} / \beta_{i,t}^{HOR}$ for the HVS and GG homeownership rates, respectively, with standard errors block bootstrapped by CBSA. The 95 percent confidence intervals are shown in bars.

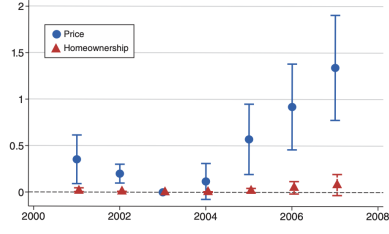


FIGURE 4. DK APL PREEMPTION REDUCED FORM

Notes: The figure shows estimates of the cumulative sum from 2003 of β_i for each indicated year and outcome variable (price or GG-Microdata homeownership rate) estimated from equation (2), with the instrument being $Z_t = APL_{2004} \times OCC_{2003}$. The controls include median income growth, population growth, the Saiz (2010) elasticity interacted with a dummy for post-2004, the fraction of loans originated by OCC-regulated lenders interacted with a dummy for post-2004, and the fraction of OCC-regulated lenders interacted with a dummy for APLs. All regressions are weighted by 2000 population, and standard errors are clustered by CBSA, as in the original DK paper. The 95 percent confidence intervals are shown in bars.

2.5 Figure 5: MS (PLS expansion) credit shock

What Figure 5 shows.

- A bank-liquidity-driven credit expansion increases p/q .
- The homeownership effect is comparatively small.

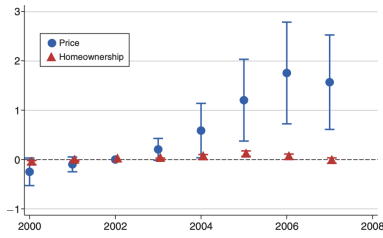


FIGURE 5. MS PLS EXPANSION REDUCED FORM

Notes: The figure shows estimates of the effect of a city's NCL share on the indicated outcome (price or homeownership rate) based on estimating equation (2) with the instrument being $Z_t = NCLShare_t^{2002}$ and 2002 being the base year. All standard errors are clustered by CBSA and all regressions are weighted by housing units as in the original MS paper. 95 percent confidence intervals are shown in bars.

TABLE 1—PARAMETER VALUES: BASELINE CALIBRATION (QUARTERLY)

Parameter	Name	Value	Internal	Target/Source
<i>Demographics and preferences</i>				
Borrower pop. share	χ_B	0.626	N	1998 SCF
Borrower inc. share	s_B	0.525	N	1998 SCF
Landlord pop. share	χ_L	0.000	N	Normalization
Borr. discount factor	β_B	0.974	Y	PMI rate (see text)
Saver discount factor	β_S	0.992	Y	Nom. interest rate = 6.46%
Landlord discount factor	β_L	0.974	Y	Equal to β_B
Housing utility weight	ξ	0.200	N	Davis and Ortalo-Magné (2011)
Saver housing demand	$\hat{H}_S$	5.299	Y	Steady state optimum
<i>Ownership benefit heterogeneity</i>				
Landlord het. (location)	$\mu_{\omega,L}$	0.861	Y	Avg. homeownership rate
Landlord het. (scale)	$\sigma_{\omega,L}$	9.336	Y	Empirical elasticities (Section IV)
Borr. het. (location)	$\mu_{\omega,B}$	0.211	Y	Borr. VTI (1998 SCF)
Borr. het. (scale)	$\sigma_{\omega,B}$	0.697	Y	Implied subsidy (see text)
<i>Technology and government</i>				
New land per period	$\bar{L}$	0.090	Y	Residential inv = 5% of GDP
Land share of construction	φ	0.629	N	Res inv. elasticity in boom
Housing depreciation	δ	0.005	N	Standard
Inflation	π	1.008	N	3.22% annualized
Tax rate	τ	0.204	N	Standard
<i>Mortgage contracts</i>				
Share of active movers	η	0.496	Y	$\hat{H}_B / \hat{H}$
Moving rate	ρ	0.034	N	Greenwald (2018)
Loan amortization	ν	0.45%	N	Greenwald (2018)
Borr. LTV Limit	θ_L^{TV}	0.850	N	Greenwald (2018)
Borr. PTI Limit	θ_B^{TV}	0.360	N	Greenwald (2018)
PTI offset (other debt)	π_c	0.08	N	Greenwald (2018)
PTI offset (taxes + insurance)	α	0.090%	N	Greenwald (2018)
Landlord LTV Limit	θ_L^{TV}	0.000	N	No landlord credit

2.6 Table 1: Baseline calibration (parameters)

Reading of Table 1.

- The model is quarterly; standard parameters govern depreciation, land share in construction, housing weight in utility, taxes, inflation.

- Borrowers face both LTV and PTI constraints with baseline limits $\theta_B^{LTV} = 0.85$ and $\theta_B^{PTI} = 0.36$.
- The key segmentation discipline is the landlord “rental suitability” dispersion parameter $\sigma_{\omega,L}$, which is chosen to match the empirical elasticities from the credit-shock impulse responses.
- Baseline rules out landlord leverage ($\theta_L^{LTV} = 0$), and later extensions relax this.

2.7 Figure 6: Matching the empirical elasticities in the model

Interpretation.

- Panels A–B: benchmark model reproduces “big p/q response, small HO response”.
- Panel C: the data-implied inverse ratio κ_k is close to zero; the model is calibrated to match that.
- Panel D: counterfactual segmentations illustrate identification: with no segmentation, HO reacts too much; with full segmentation, HO barely reacts.

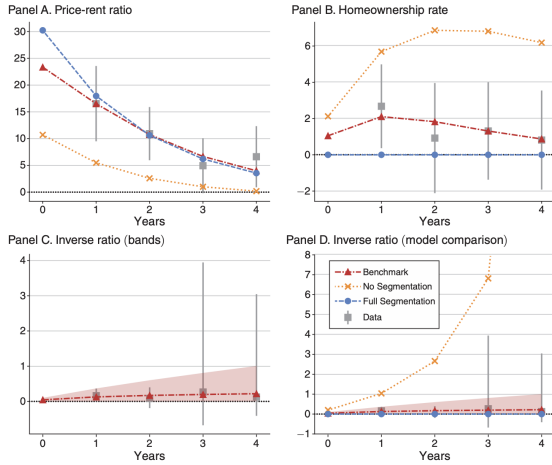


FIGURE 6. IMPULSE RESPONSES: MODEL VERSUS DATA

Notes: Gray squares and bands indicate our benchmark LS estimates from Figure 3 and their 95 percent confidence intervals. We omit Year 0 estimates, which are not used in the calibration exercise. These empirical estimates are plotted alongside the corresponding outputs of each model. For each model, we compute a quarterly impulse response, then average over each year to obtain annual responses. Panels A and B correspond to panel A of Figure 3, while panels C and D correspond to panel C of Figure 3. The shaded red area indicates the range of outcomes from the lower bound estimate of $\sigma_{\omega,L} = 1.594$, to the upper bound estimate of $\sigma_{\omega,L} = \infty$, equivalent to the Full Segmentation case.

TABLE 2—RESULTS, BOOM EXPERIMENTS

Experiment	Price-rent	Homeown.	Loan-inc.
Peak data increase	51.5%	3.3 pp	71.2%
<i>Credit relaxation (share of peak data increase)</i>			
Full segmentation	36%	0%	54%
Benchmark	32%	17%	52%
Est. lower bound	22%	71%	46%
No segmentation	−2%	180%	31%
<i>Credit relaxation + decline in rates (share of peak data increase)</i>			
Full segmentation	77%	0%	88%
Benchmark	70%	35%	83%
Est. lower bound	47%	136%	69%
No segmentation	2%	307%	36%
<i>Removing credit relaxation from full boom (share of peak data increase)</i>			
Full boom (benchmark)	100%	100%	107%
No credit relaxation (benchmark)	47%	72%	29%
Full boom (no segmentation)	100%	100%	114%
No credit relaxation (no segmentation)	103%	−297%	45%
<i>Credit relaxation + decline in rates: Extensions (share of peak data increase)</i>			
Landlord credit	70%	27%	79%
Saver demand	50%	17%	103%

Notes: This table summarizes the results from the various nonlinear transition experiments in Sections V and VI. “Price-rent” is the price-rent ratio, “Homeown.” is the homeownership rate, and “Loan-Inc.” is the aggregate loan to income ratio. The top row displays the actual changes in these variables, in levels from 1998:1 to the peak of each series during the boom period (2006–2008). The remaining numbers below display the shares of these peak increases explained by each model-experiment combination, calculated from 1998:1 to the peak of each model boom in 2007:1. The loan-to-income ratio is the ratio of household residential mortgages to household gross income (Federal Reserve Board of Governors 2021). For other data definitions, see the notes for Figure 1.

2.8 Table 2: Boom accounting (what fraction of the peak is explained?)

Key numbers I would quote (benchmark column).

- *Credit relaxation alone* explains about **32%** of the peak p/q increase, **17%** of the peak HO increase, and **52%** of the peak loan-to-income increase.
- *Credit relaxation + falling rates* explains about **70%** of peak p/q , **35%** of peak HO, and **83%** of peak loan-to-income.

Cross-segmentation comparison (mechanism check).

- Full segmentation: credit mostly shows up in p/q , essentially none in HO.
- No segmentation: credit shows up *too much* in HO and too little in p/q (even wrong-signed p/q in the credit-only experiment).

2.9 Figure 7: Credit relaxation experiment in the model

Interpretation.

- Panel A: credit standards alone move leverage and (depending on segmentation) p/q , but typically do not match the full boom.
- Panel B: adding the observed decline in mortgage spreads/rates substantially improves fit.
- The segmentation parameter determines whether the model matches the *small* increase in HO in the data.

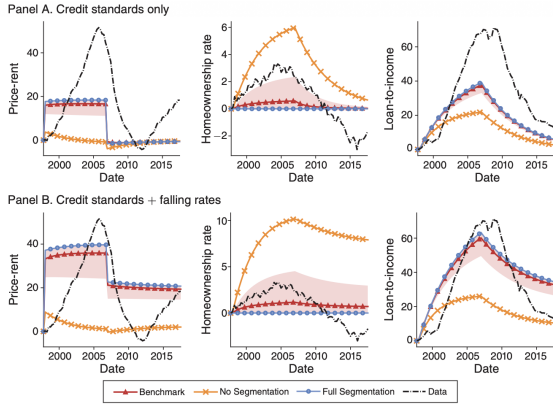


FIGURE 7. CREDIT RELAXATION EXPERIMENT

Notes: Each panel displays perfect foresight paths following a relaxation of LTV and PTI constraints. The “Benchmark” model sets a value of $\sigma_{u,L}$ calibrated to match our empirical IRFs as in Section IV, while the “No Segmentation” model sets $\sigma_{u,L} = 0$ and the “Full Segmentation” model sets $\sigma_{u,L} \rightarrow \infty$. Shaded bands indicate the range of outcomes from the lower bound estimate of $\sigma_{u,L} = 1.594$, to the upper bound estimate of $\sigma_{u,L} = \infty$, equivalent to the Full Segmentation case. Results are summarized numerically in Table 2. For data definitions, see notes for Figure 1 and Table 2.

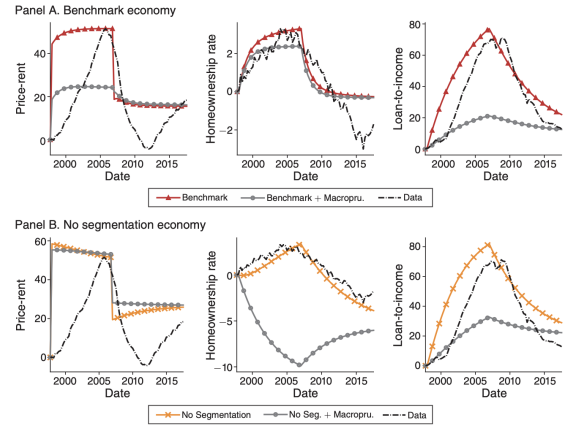


FIGURE 8. FULL BOOM EXPERIMENT

Notes: Plots display perfect foresight paths. The “Benchmark” model calibrates $\sigma_{u,L}$ to match our empirical IRFs as in Section IV, while the “No Segmentation” model sets $\sigma_{u,L} = 0$. For each set of plots, the colored plots display an experiment imposing a credit relaxation, decline in the interest rate, and level shifts to the demand and supply curves ($\mu_{u,B}$ and $\mu_{u,L}$), with these level shifts chosen to exactly match the peak growth of price-rent ratios and the homeownership rate over the housing boom. In each panel, the responses ending in “+ Macropru.” display an alternative experiment showing the same decline in the mortgage rate and level shifts to our demand and supply curves while removing the relaxation of credit standards. Results are summarized numerically in Table 2. For data definitions, see notes for Figure 1 and Table 2.

2.10 Figure 8: Full boom + macroprudential counterfactual (no credit relaxation)

Interpretation.

- “Full boom” introduces additional shocks so the model matches the full boom-bust path.
- “Macropru” removes credit relaxation (keeping other shocks) to measure how much of the boom is due to credit standards.

- In the benchmark (segmented) economy, the macroprudential counterfactual meaningfully attenuates the boom in p/q and leverage.
- In the no-segmentation economy, the counterfactual dynamics for HO and prices are implausible, emphasizing why tenure supply must be disciplined by the data.

3 Empirical models (regressions) used in the paper

3.1 Local projection with LS (CLL-based) instrument

For each horizon k (in years), estimate:

$$\log Y_{i,t+k} = \xi_i + \psi_t + \beta_k Z_{i,t} + \theta' X_{i,t} + \varepsilon_{i,t}, \quad (1)$$

where $Y_{i,t+k} \in \{p/q, \text{HO}\}$, ξ_i are CBSA fixed effects, ψ_t are time fixed effects.

Instrument:

$$Z_{i,t}^{\text{LS}} = \text{ShareNearCLL}_{i,t-1} \times \% \Delta \text{CLL}_t. \quad (2)$$

3.2 Event-study style specification for DK (APL preemption)

$$\log Y_{i,t} = \xi_i + \psi_t + \sum_{k \neq \tau} \beta_k (\text{APL}_i \times \text{OCCShare}_i) \mathbf{1}\{t = k\} + \theta' X_{i,t} + \varepsilon_{i,t}. \quad (3)$$

3.3 Inverse ratio summary for tenure supply

Let $\beta_k^{p/q}$ be the estimated response of $\log(p/q)$ and β_k^{HO} be the response of HO. Define

$$\kappa_k \equiv \frac{\beta_k^{\text{HO}}}{\beta_k^{p/q}}. \quad (4)$$

Interpretation. A small κ_k means HO barely responds when p/q moves a lot. This implies a steep tenure supply curve.

4 Structural model: all key equations and objects

4.1 Preferences

Borrowers and savers have log utility over a Cobb–Douglas aggregator of nondurables c and housing services h :

$$U_j = \sum_{t=0}^{\infty} \beta_j^t \log(c_{j,t}^{1-\xi} h_{j,t}^{\xi}), \quad j \in \{B, S\}. \quad (5)$$

Landlords are risk-neutral:

$$U_L = \sum_{t=0}^{\infty} \beta_L^t c_{L,t}. \quad (6)$$

4.2 Borrower mortgage constraints (static)

Active borrowers choose new mortgage $M_{i,t}^*$ and house $H_{i,t}^*$ subject to

$$M_{i,t}^* \leq \theta_B^{\text{LTV}} p_t H_{i,t}^*, \quad (7)$$

$$M_{i,t}^* \leq \frac{(\theta_B^{\text{PTI}} - \kappa) y_{B,t} e_{i,t}}{r_{B,t}^* + \nu + \alpha}. \quad (8)$$

4.3 Borrower budget constraint (flow)

In aggregate, borrower resources satisfy

$$\begin{aligned} c_{B,t} \leq & (1 - \tau) y_{B,t} + \rho \eta (M_{B,t}^* - \pi^{-1} (1 - \nu) M_{B,t-1}) - \pi^{-1} (1 - \tau) X_{B,t-1} - \nu \pi^{-1} M_{B,t-1} \\ & - \rho p_t (\eta H_{B,t}^* - H_{B,t-1}) - \delta p_t H_{B,t-1} - q_t (h_{B,t} - H_{B,t-1}) + \left(\int_{\bar{\omega}_{B,t}}^{\infty} \omega d\Gamma_{\omega,B} \right) q_t \hat{H}_{t-1} + T_{B,t}. \end{aligned} \quad (9)$$

4.4 Mortgage and housing state evolution

With moving probability ρ and amortization rate ν :

$$M_{B,t} = \rho \eta M_{B,t}^* + (1 - \rho) \frac{1 - \nu}{\pi} M_{B,t-1}, \quad (10)$$

$$X_{B,t} = \rho \eta r_{B,t}^* M_{B,t}^* + (1 - \rho) \frac{1 - \nu}{\pi} X_{B,t-1}, \quad (11)$$

$$H_{B,t} = \rho \eta H_{B,t}^* + (1 - \rho) H_{B,t-1}. \quad (12)$$

4.5 Tenure market clearing with cutoffs

Cutoffs are pinned by the homeownership share:

$$1 - \Gamma_{\omega,B}(\bar{\omega}_{B,t}) = \frac{H_{B,t}}{\hat{H}_t}, \quad \Gamma_{\omega,L}(\bar{\omega}_{L,t}) = \frac{H_{B,t}}{\hat{H}_t}. \quad (13)$$

4.6 Construction and housing supply

Aggregate housing stock evolves as

$$\hat{H}_t = (1 - \delta) \hat{H}_{t-1} + I_t, \quad I_t = L_t^\varphi Z_t^{1-\varphi}, \quad (14)$$

where L_t is land input and Z_t is a numeraire input. Land supply is capped each period (a permit/land constraint).

4.7 Key asset-pricing representations: tenure demand and tenure supply

The equilibrium can be summarized by two schedules (paper’s main reduced forms):

$$p_t^{\text{Demand}}(H_{B,t}) = \frac{\mathbb{E}_t \left[\Lambda_{B,t+1} \left((1 + \bar{\omega}_{B,t}) q_{t+1} + (1 - \delta - (1 - \rho) C_{B,t+1}) p_{t+1} \right) \right]}{1 - C_{B,t}}, \quad (15)$$

$$p_t^{\text{Supply}}(H_{B,t}) = \mathbb{E}_t \left[\Lambda_{L,t+1} \left((1 + \bar{\omega}_{L,t}) q_{t+1} + (1 - \delta) p_{t+1} \right) \right]. \quad (16)$$

What shifts demand? Credit conditions raise $C_{B,t}$ (the collateral value of housing), lowering the effective downpayment wedge $1 - C_{B,t}$ and shifting demand out.

What makes supply steep? The landlord cutoff $\bar{\omega}_{L,t}$ depends on HO. If $\bar{\omega}_{L,t}$ is very sensitive to HO, then p/q must change a lot to induce tenure reallocation.

5 Calibration of segmentation

5.1 Heterogeneity distributions

The paper parameterizes borrower and landlord surplus using logistic CDFs:

$$\Gamma_{\omega,j}(\omega) = \left(1 + \exp(-(\omega - \mu_{\omega,j})/\sigma_{\omega,j}) \right)^{-1}, \quad j \in \{B, L\}. \quad (17)$$

Segmentation parameter. $\sigma_{\omega,L}$ controls dispersion in rental suitability and therefore the steepness of tenure supply.

5.2 How it is disciplined

They pick $\sigma_{\omega,L}$ (and the credit-shock process parameters) to match the empirical impulse responses of $\log(p/q)$ and HO to a credit shock. The inverse ratio κ_k is the main diagnostic.

6 Boom experiments (model inputs)

- **Credit relaxation path (standards).** In the boom experiment, the paper relaxes constraints roughly from baseline

$$\theta_B^{\text{LTV}} : 0.85 \rightarrow 0.99, \quad \theta_B^{\text{PTI}} : 0.36 \rightarrow 0.65,$$

starting around 1998Q2 and reversing in 2007Q1.

- **Mortgage-rate decline.** Add a roughly 2 percentage point decline in mortgage spreads/rates (treated as a permanent shift over the sample window).
- **Full boom exercise.** Add extra shocks (demand/supply) so the model matches the full boom-bust; then remove credit relaxation to quantify a macroprudential counterfactual.

7 Bottom line (how I would summarize the paper in a few sentences)

- Credit conditions *do* move house prices (and price–rent ratios), but the channel is not primarily “massively higher homeownership.”
- Empirically, the homeownership response to credit-supply shocks is small relative to the price response, implying a steep tenure supply curve (segmented rental market).
- A structural model disciplined by this tenure elasticity implies that easing credit standards (and falling mortgage rates) can explain a large fraction of the 2000s rise in prices and leverage while matching the modest homeownership movement.

8 Conclusion

8.1 Summary

- **Why the literature disagreed.** Existing papers span conclusions from “credit barely matters” to “credit explains most of the boom” because they implicitly assume different *degrees of housing-market segmentation* between *credit-sensitive* households (marginal owner-occupiers) and *credit-insensitive* agents who can absorb demand (landlords and/or unconstrained savers).
- **Empirical sufficient statistic for segmentation.** The paper shows that the *relative* causal effect of credit on the *price–rent ratio* versus the *homeownership rate* is highly informative about the slope of the tenure supply curve:

$$\kappa \equiv \frac{\partial \text{HO} / \partial Z}{\partial \log(p/q) / \partial Z}.$$

Empirically, across multiple credit-supply instruments, $\partial \log(p/q) / \partial Z$ is large while $\partial \text{HO} / \partial Z$ is muted and often statistically insignificant, implying **small** κ and therefore a **steep tenure supply curve**.

- **Quantitative bottom line.** In a structural model calibrated to match these elasticities, the implied degree of segmentation is *close to the perfect-segmentation benchmark*, i.e., rental-market frictions are large. Under that calibration, the relaxation of credit standards can explain a **large fraction** of the 2000s boom in valuation:
 - In the boom-accounting exercise, *credit relaxation alone* explains about **32%** of the peak rise in p/q (benchmark column of Table 2).
 - In the full-boom counterfactual, preventing credit standards from relaxing reduces the observed boom in p/q by about **53%** (macroprudential counterfactual logic).
 - When *falling mortgage rates* are combined with credit relaxation, the model can explain an even larger share of peak p/q movements (Table 2 reports roughly **70%** in the benchmark calibration).

- **Conceptual implication.** The paper’s key message is not merely that “credit matters,” but that **credit can move house prices substantially even when homeownership barely changes**. The reason is that, with steep tenure supply, demand pressure is capitalized into prices rather than translated into large tenure reallocation.
- **Robustness via extensions.** Allowing landlords to use credit or allowing savers to adjust housing demand can dampen or re-shape the responses, but the paper argues the core takeaway remains: empirically disciplined segmentation is strong enough that credit supply still explains a *meaningful* portion of the boom in p/q .

8.2 Economic intuition: one coherent way to explain it in words

- **Start from an equilibrium mapping.** Think of the housing market equilibrium in the $(HO, p/q)$ plane. A credit easing shifts *tenure demand* outward (more households can finance ownership).
- **Who absorbs the extra demand determines what moves.** The crucial question is whether there exist sufficiently elastic, credit-insensitive buyers of housing services/units (landlords and/or savers) who can absorb the extra demand without large price changes.
 1. If absorption is easy (flat tenure supply; little segmentation), the main adjustment is a higher homeownership rate: HO rises a lot, p/q rises little.
 2. If absorption is hard (steep tenure supply; strong segmentation), the main adjustment is valuation: p/q rises a lot, while HO rises only modestly.
- **Why the price–rent ratio is the right valuation object.** p/q removes level shifts in rents and isolates how the market capitalizes future housing services. If credit mainly affects *who holds the house* (tenure) but not overall scarcity/valuation, then p/q should not move much. The paper finds the opposite: p/q moves strongly.
- **Why the “ratio of elasticities” is a sufficient statistic.** The empirical object κ tells you how much tenure reallocation you get per unit of valuation response. When κ is near zero, the market is behaving as if tenure supply is close to vertical: the marginal credit-driven demand is priced in rather than reallocated into a large change in ownership.
- **The unifying punchline.** Once segmentation is measured rather than assumed, the apparently conflicting conclusions in the literature become different points on the same spectrum. In the empirically relevant region of that spectrum (steep tenure supply), credit standards are a quantitatively important driver of house-price valuation dynamics in the 2000s.

8.3 Takeaway

Greenwald and Guren (2025) show that easier credit conditions can raise house prices substantially even when homeownership changes little, because with a steep and segmented tenure supply curve, credit shocks are capitalized mainly into the price–rent ratio rather than broad tenure reallocation.